

DATA STRUCTURES STUDY MATERIAL

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Data Structure

DATA STRUCTURES :

Data Structure is a particular way of storing and organizing data in a computer's memory so that the data can be efficiently accessed.

What is the need of DATA STRUCTURE?

If we make any program using suitable Data Structure then it will be efficient in terms of Time and Space complexity.

What are the various types of Data Structure ?

There are two types of Data Structure :

1. Primitive Data Structure.
2. Non-Primitive Data Structure.



PRIMITIVE DATA STRUCTURE:

Primitive Data Structures are those Data Structures which can store only single value.

Example : int, float, bool, char, etc .

NON - PRIMITIVE DATA STRUCTURE:

Non-Primitive Data Structure are those Data Structures which are derived from Primitive Data Structures.

They are further categorized into two types:

1. Linear Data Structure.
2. Non-Linear Data Structure.



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LINEAR DATA STRUCTURE :

Those Data Structure in which one element is only connected to one other element are called Linear Data Structure.

Example : Array, string, Stack, Queue, linked list.

NON - LINEAR DATA STRUCTURE:

Those Data Structure in which one element is connected to multiple element are called Non-Linear Data Structure.

Example : Tree, Graph.



ALGORITHM:

It is a step by step sequence to solve any problem, before writing any program, We first write various algorithm then we analysis those algorithms on the basis of time and space complexity. The algorithm which suits best approach according to time and space is implemented using a suitable programming language.



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ASYMPTOTIC NOTATION:

These are the notations which determine the performance of an algorithm in time and space complexity.

There are Three types of Asymptotic Notation :

1. Best Case (Omega, Ω)
2. Worst Case (Big Oh, O)
3. Average Case (Theta, θ)



CALCULATE TIME COMPLEXITY

```
for(i=0;i<n;i++)  
{  
    print("Sai Campus Recruitment Training")  
}
```

CALCULATE TIME COMPLEXITY

$O(N)$



CALCULATE TIME COMPLEXITY

```
for(i=0;i<n/2;i++)  
{  
    print("Sai Campus Recruitment Training")  
}
```


CALCULATE TIME COMPLEXITY

$O(N)$

CALCULATE TIME COMPLEXITY

```
for(i=1;(i^2)<=n;i++)  
{  
    print("Sai Campus Recruitment Training")  
}
```



CALCULATE TIME COMPLEXITY

$O(\text{Root } n)$

CALCULATE TIME COMPLEXITY

```
for(i=1;i<=n;i=i*2)
{
    print("Sai Campus Recruitment Training")
}
```

CALCULATE TIME COMPLEXITY

$O(\log n)$



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CALCULATE TIME COMPLEXITY

```
for(i=0;i<n;i++)  
{  
    for(j=0;j<n;j++)  
    {  
        print("Sai Campus Recruitment Training")  
    }  
}
```



CALCULATE TIME COMPLEXITY

$O(N^2)$

CALCULATE TIME COMPLEXITY



DSA NOTES

```
for(i=0;i<n;i++)
```

```
{
```

```
    for(j=0;j<n;j++)
```

```
    {
```

```
        for(k=0;k<n;k++)
```

```
        {
```

```
            print("Sai Campus Recruitment Training")
```

```
        }
```

```
    }
```

```
}
```



CALCULATE TIME COMPLEXITY

$O(N^3)$

CALCULATE TIME COMPLEXITY



DSA NOTES

```
for(i=0;i<n;i++)
{
    for(j=0;j<n;j=j*2)
    {
        print("Sai Campus Recruitment Training")
    }
}
```



CALCULATE TIME COMPLEXITY

$O(n \log n)$

```
for(i=1;i<=n;i++)  
{  
    sum=sum+i  
}  
print(sum)
```



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CALCULATE TIME COMPLEXITY

$O(n)$

n=somevalue

sum=n*(n+1)/2

print(sum)



CALCULATE TIME COMPLEXITY

$O(1)$



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CALCULATE TIME COMPLEXITY



DSA NOTES

a=5

b=6

a=a+b

b=a-b

a=a-b

print(a,b)



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CALCULATE TIME COMPLEXITY

$O(1)$

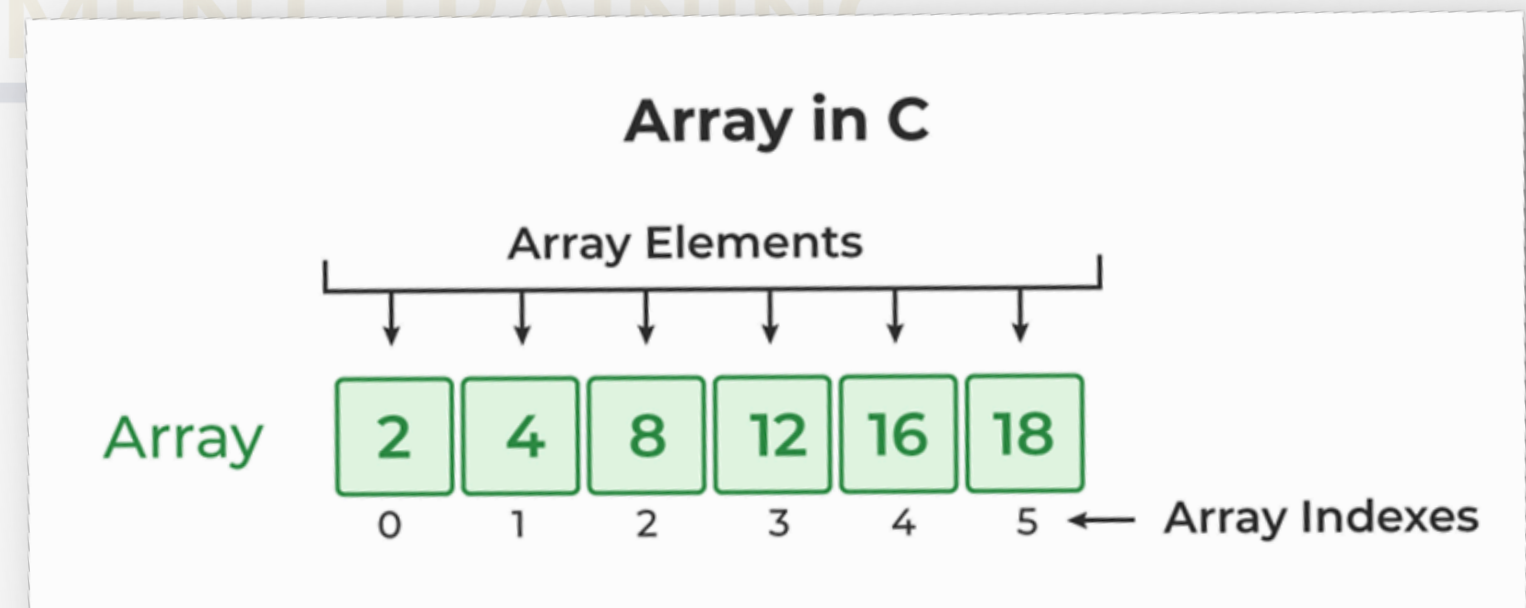
What is Array?

Array is the collection of similar datatype stored in continuous memory location.

In Array, Index always start with 0 and end with n-1 .

Syntax of Array :

Datatype arrayname[size];



Advantage of Array:

Accessing: We can access any element of array in constant time with $O(1)$ time complexity by using the formula,

Formula : Required address = Base address + Index Size.

Disadvantages of Array:

1. Static Memory Allocation: Array has Static Memory Allocation it means its size cannot be increased or decreased at runtime .
2. Insertion or Deletion: Insertion and deletion are costly and time consuming operations in Array because to “insert” any element in the array we have to shift all the elements after it one position right and to “delete” any element from the array we have to shift all the elements after it one position left.



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What is Stack ?

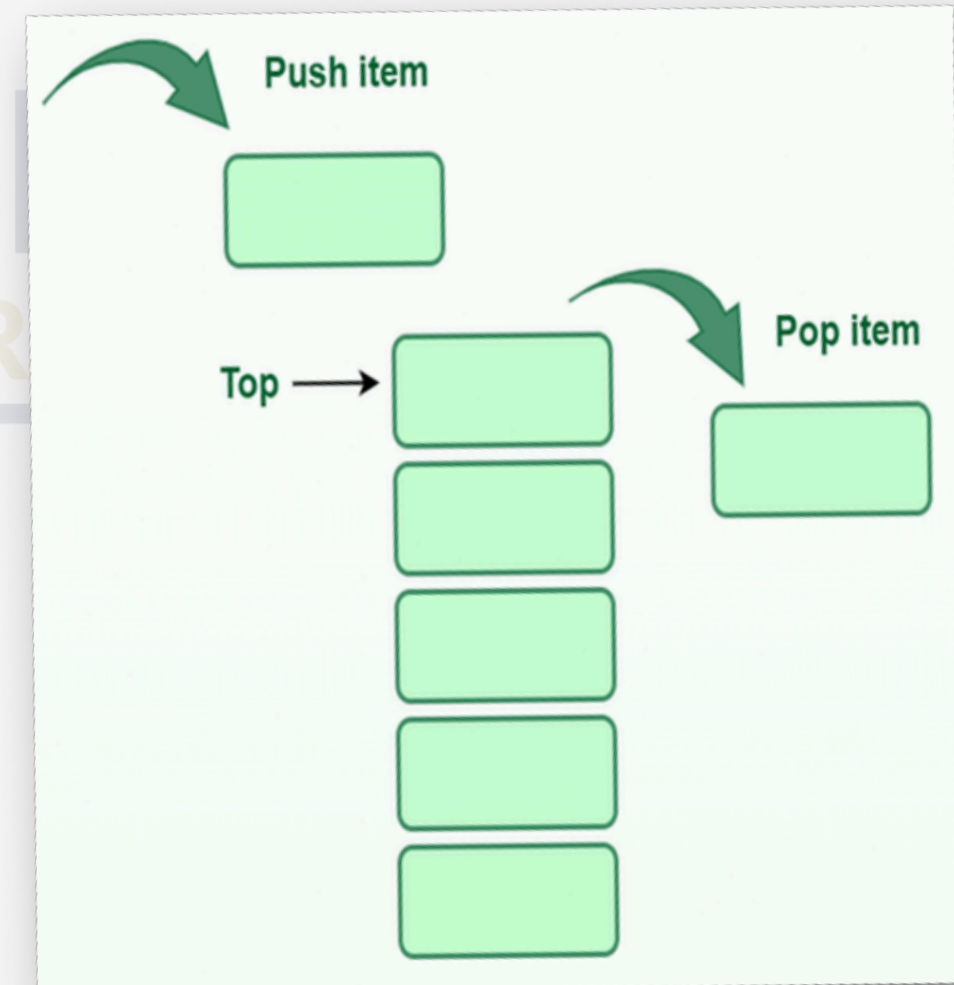
STACK is a Linear Data Structure or linear list with the restriction that elements will be inserted and deleted from only one end of the stack called top of the stack .

Stack is based on the principle of LIFO (last in first out) .

Operations on Stack :

There are three operations associated with the Stack:

1. PUSH : To insert the element into stack .
2. POP : To delete the element into stack .
3. PEEK : To see the Topmost element in the stack .



Algorithm for PUSH().

- Algorithm PUSH(arr, top, n, data)

Step-1: Check for Overflow,

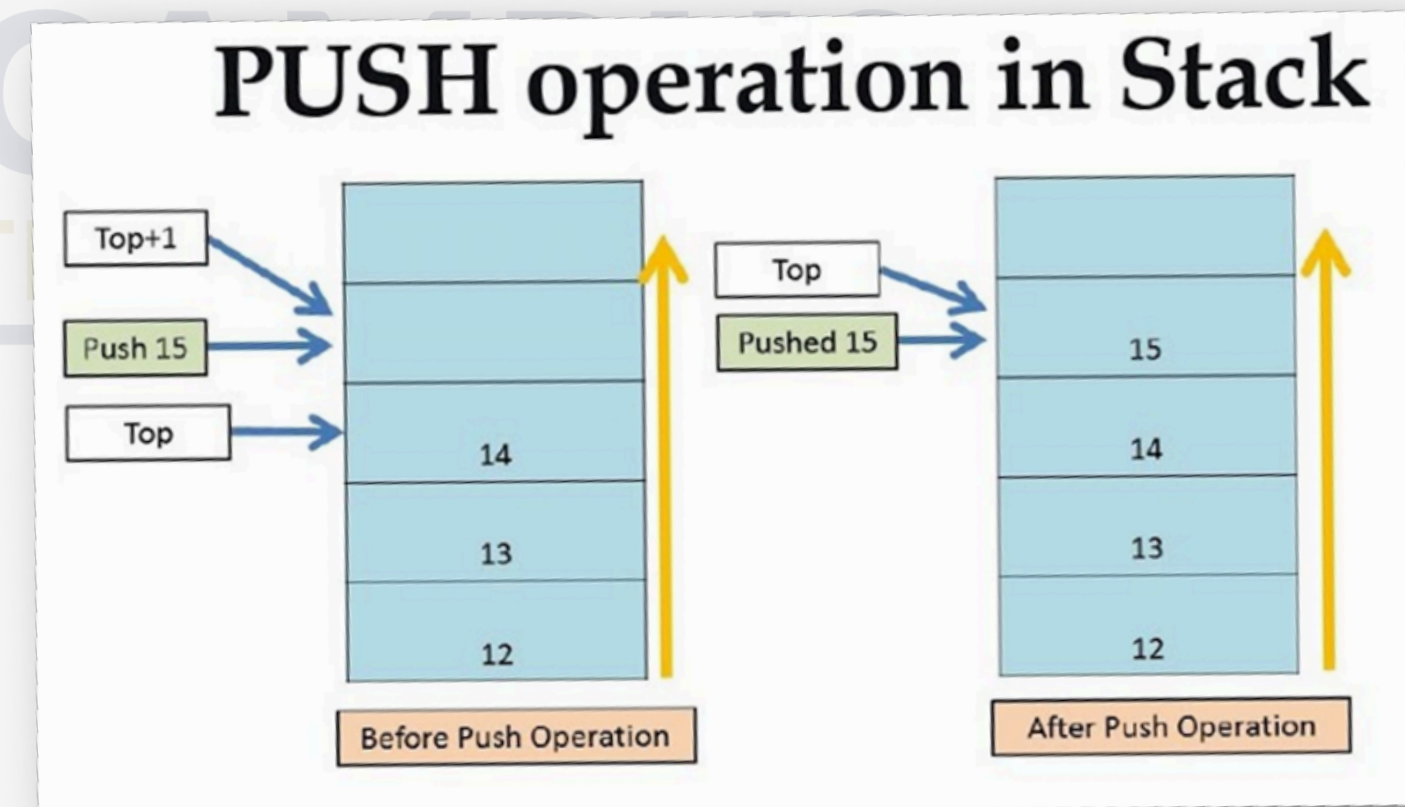
if (top == n):

print("Stack Overflow")

Exit

Step-2: arr[top] = data

Step-3: top ++



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Algorithm for POP().

- Algorithm POP(arr, top, data)

Step-1 Check for Underflow,

if (top == 0):

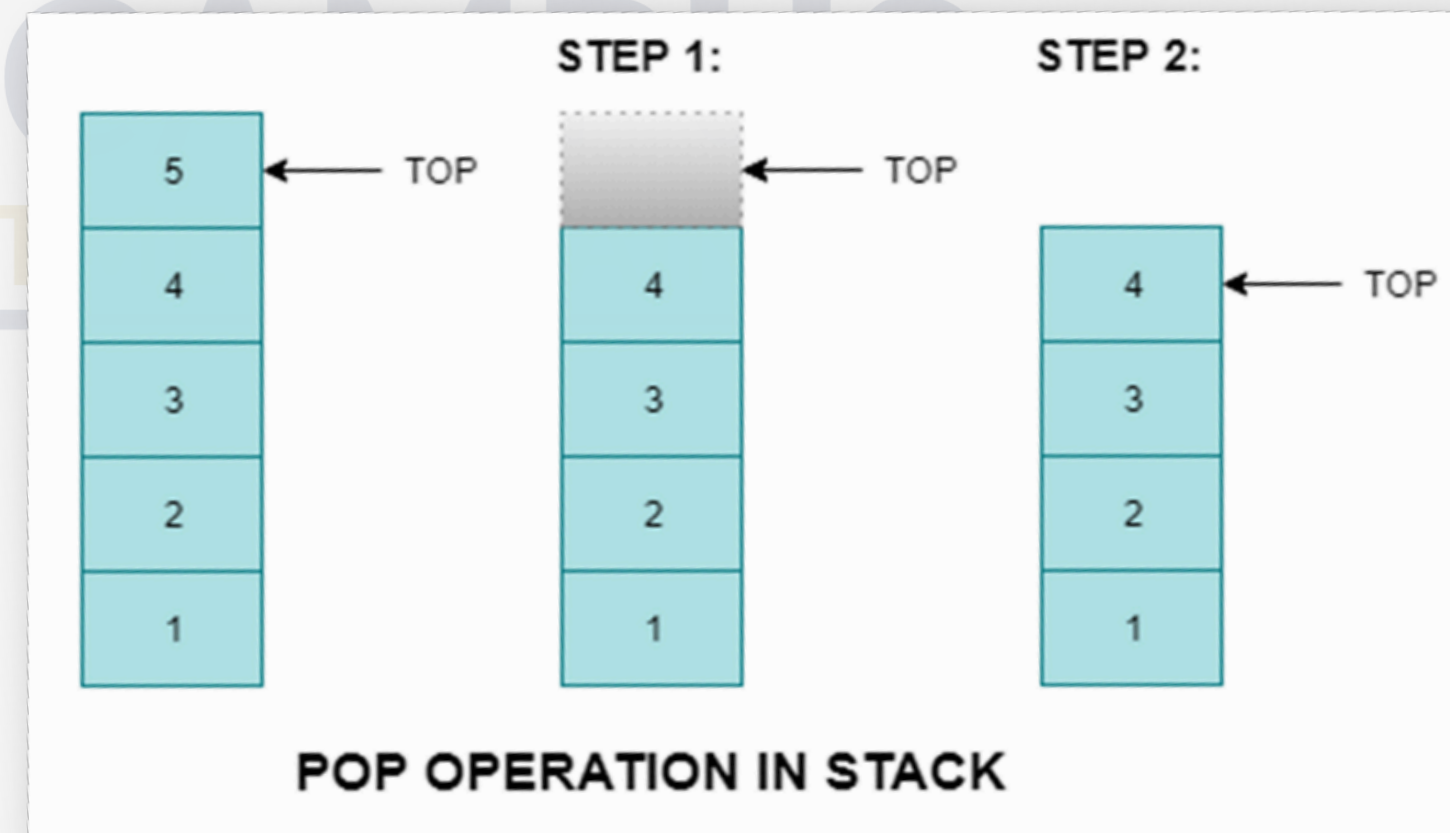
print("Stack Underflow")

Exit

Step-2: data = arr[top-1]

Step-3: top - -

Step-4: return data



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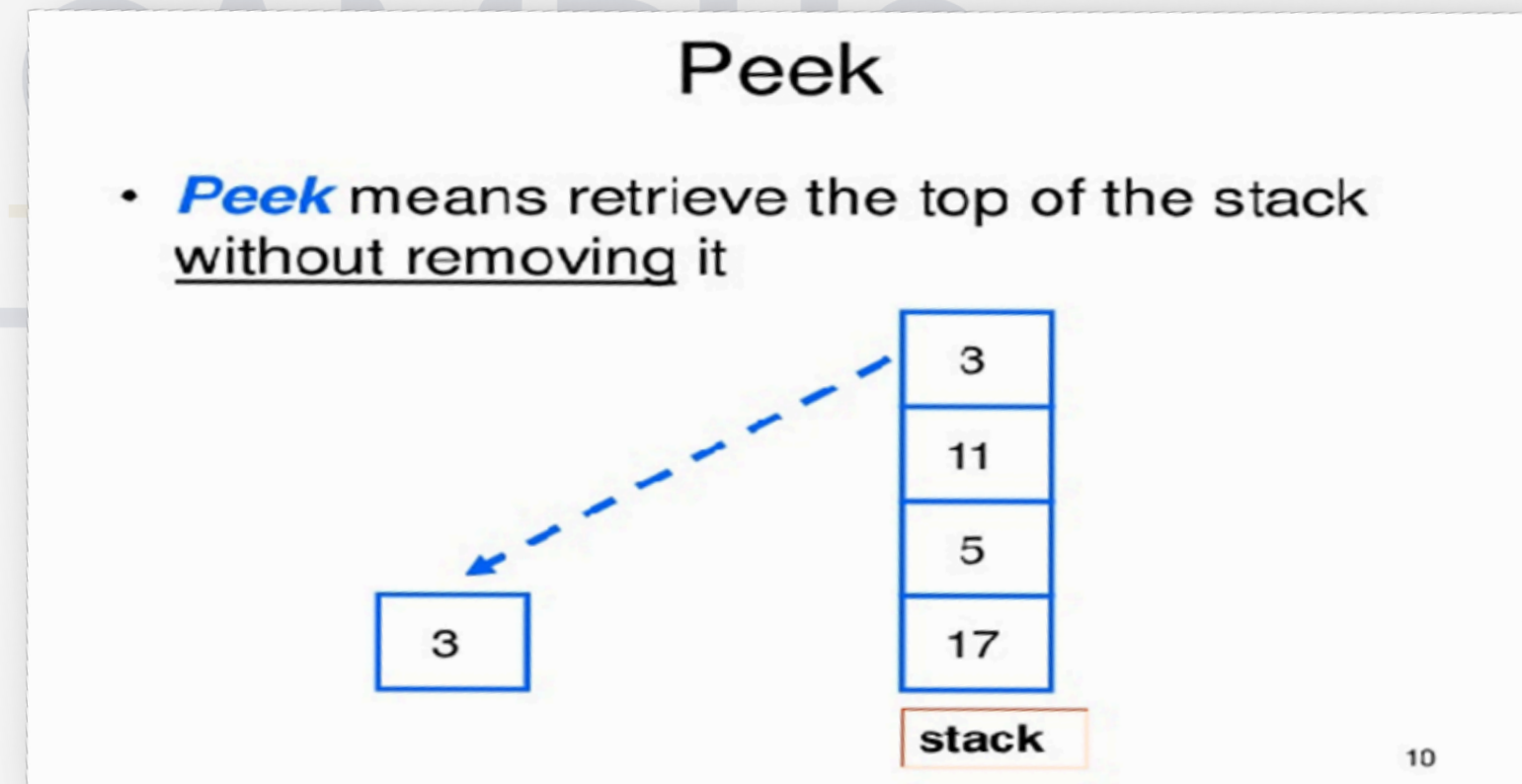
Algorithm for PEEK()

- Algorithm PEEK(arr, top, data)

Step-1: Check for Underflow,
if (top == 0):
 print("Stack Underflow")

Step-2: data = arr[top-1]

Step-3: return data



Applications of Stack:

1. To check Palindrome String: Stack can be used to check whether a string is palindrome or not. We have to PUSH all the elements of the string into the stack one by one and then POP them. If in both the cases the string remains the same then it is said to be a Palindrome String.
2. Balancing of Parenthesis: When any opening bracket comes it is Pushed into the stack and when any closing bracket comes the topmost bracket from the stack is Popped at the last if the stack is empty then Parenthesis is Balanced.
3. Function Call: All the function calls in a program are managed by stack that is the whole control mechanism is dependent on stack whenever any function is called it is always pushed into the stack.
4. Evaluation of Postfix Expression.



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What is Queue ?

Queue is a linear list with restriction that elements will be insert from one end called Rear end and delete from another end called Front end of the queue.

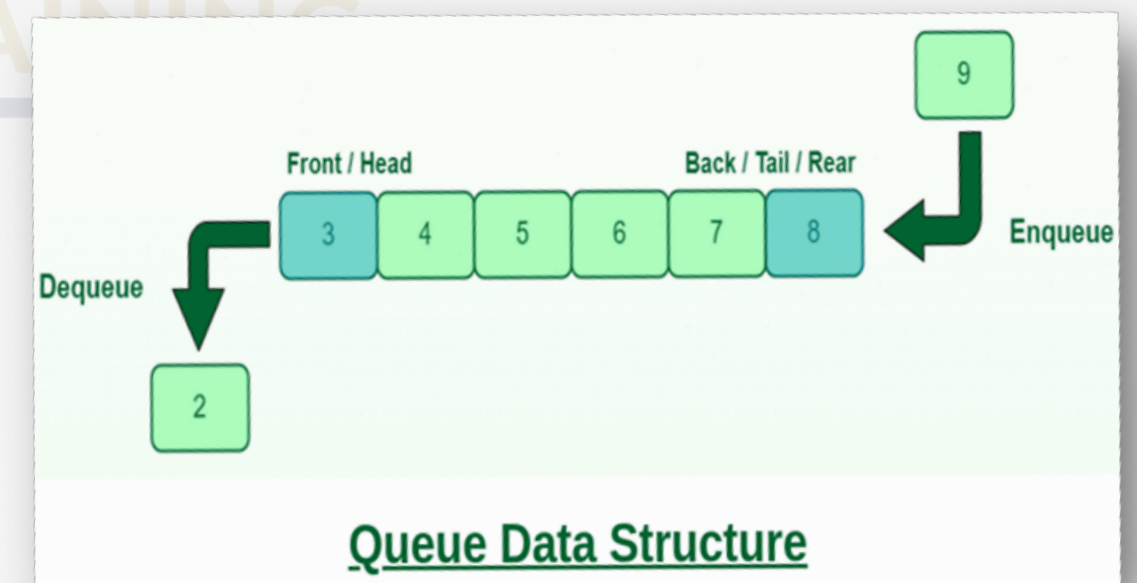
Queue is based on the principle of FIFO (first in first out).

Real-life Example: Railway ticket counter, Temple queue etc...

Technical Example: Printer, CPU etc...

There are two operations associated with Queue :

1. Enqueue: It is used to insert the element into the queue.
2. Dequeue: It is used to delete the element from the queue.



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Algorithm of Enqueue:

Algorithm Enqueue(arr, rear, n, data)

Step-1: Check for Overflow

if (rear == n):

 print("Queue Overflow")

Exit

Step-2: arr[rear] = data

Step-3: rear ++

Algorithm of Dequeue:

Algorithm Dequeue(arr, rear, front, data)

Step-1: Check for Underflow,
if (front == rear):
 print("Queue Underflow")

Step-2: data = arr[front]

Step-3: for i = 1 to rear-1:
 arr [i-1] = arr [i]

Step-4: rear - -

Step-5: return data

Types of Queue :

There are three types of queue :

1. Simple Queue: Queue is a linear list with restriction that elements will be insert from one end called Rear end and delete from another end called Front end of the queue.
2. Circular Queue: Circular Queue is a special type of queue in which the last element connected to the first element forming a ring like structure, it is called as ring buffer.
3. Priority Queue: Priority Queue is a special queue in which each element is associated with a priority and the element with higher priority are always deleted first.



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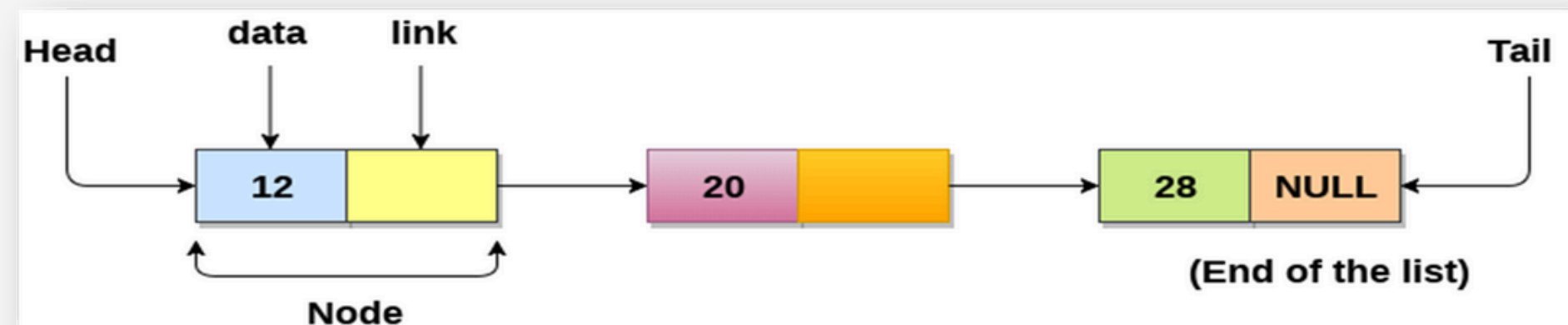
What is Linked List ?

Linked list is a collection of data in which each node contains two fields, one is data field and another is linked field/address field.

Data field contains the data and link field contains the address of next node.

Linked list has random memory allocation so a node can be present anywhere in the memory without any fixed sequence.

The address of the first node is stored in a special pointer generally called Start/Root/Head.



Types of LINKED LIST:

There are Four types of linked list:

1. Singly Linked List.
2. Singly Circular Linked List.
3. Doubly Linked List.
4. Doubly Circular Linked List.

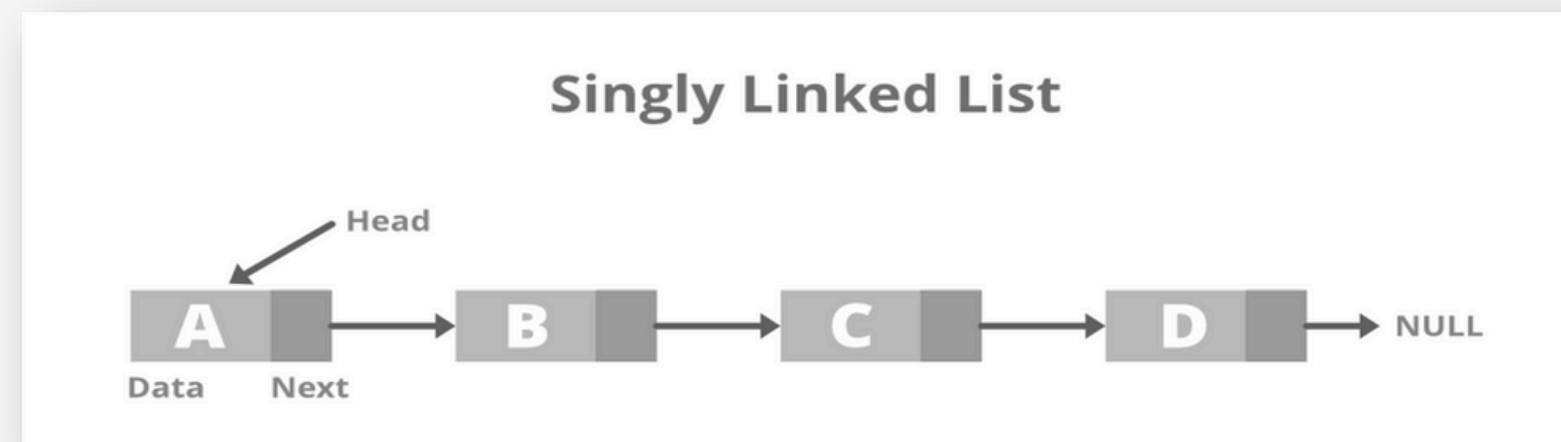
Singly Linked List :

Singly linked list is a collection of data in which each node contain two field, one is data field and another is linked field/address field.

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Singly Circular Linked List:

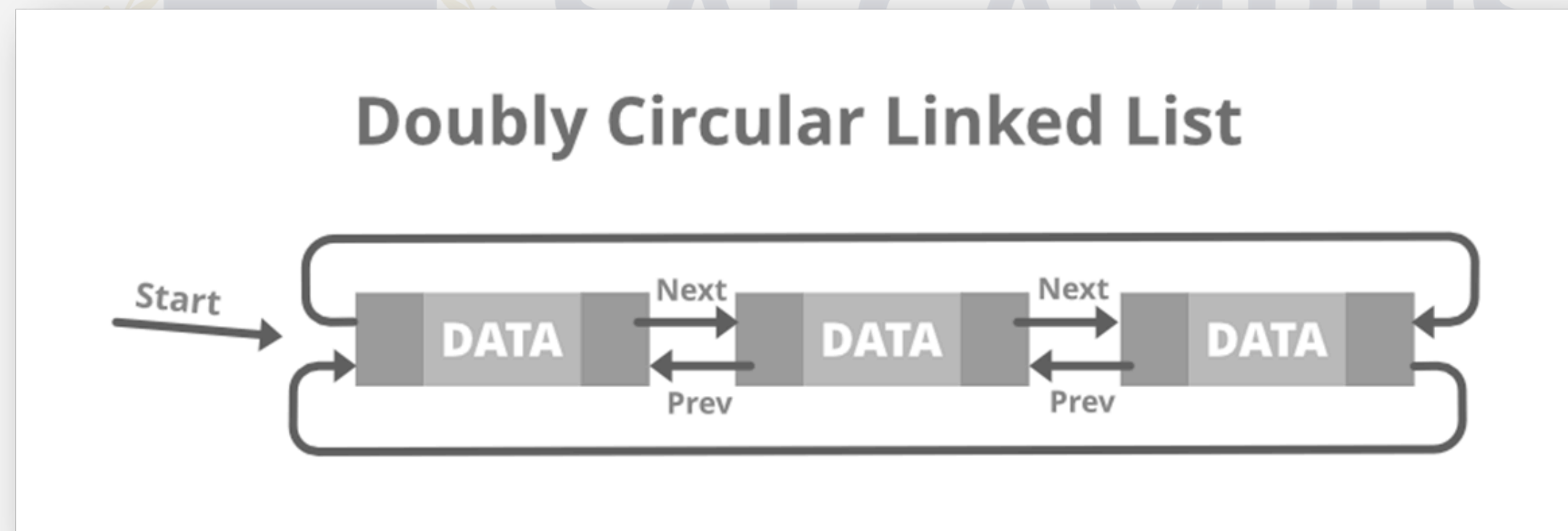
In this type of linked list the address of last node contain the address of first node.

Circular Linked List



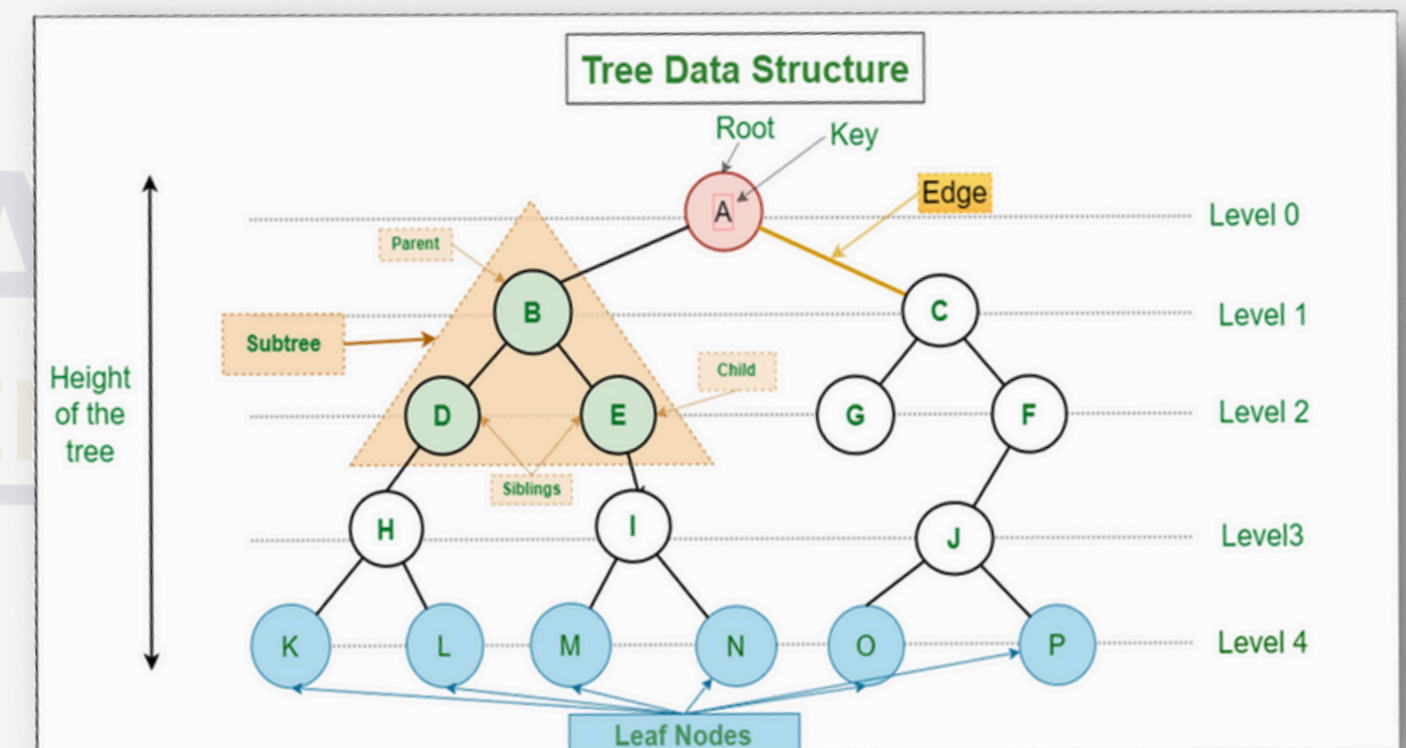
Doubly Circular Linked List :

In this linked list the r-link of last node contains the address of first node and l-link of first node contains the address of the last node.



What is Tree ?

Tree is the collection of nodes and edges. The topmost node is called as the root node. Other nodes are called as the child node. The nodes on a same level are called sibling nodes. And the nodes which do not have any child are called leaf node.



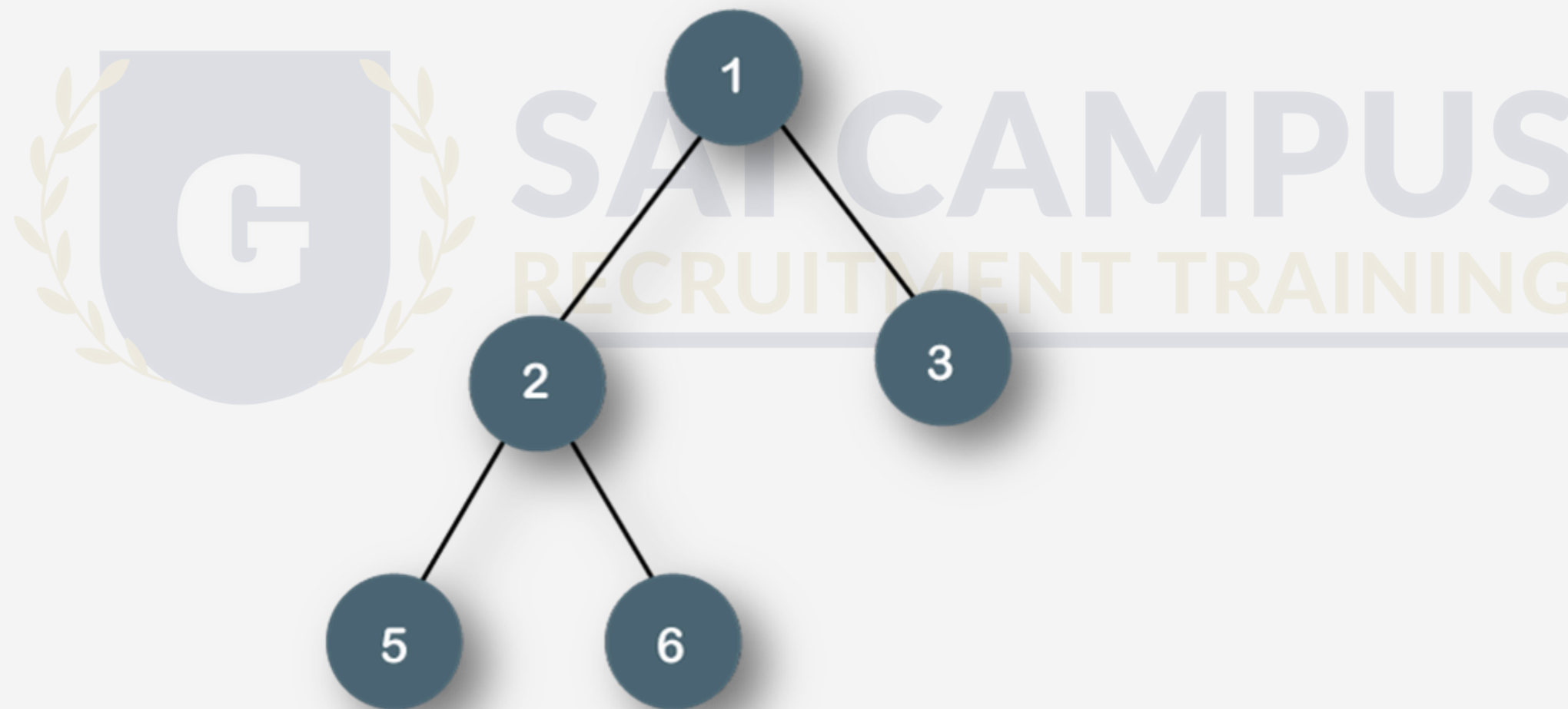
Types of TREE :

There are six types of Trees :

1. Binary Tree
2. Perfect Binary Tree
3. Strict Binary Tree
4. Complete Binary Tree
5. Binary Search Tree
6. AVL Tree

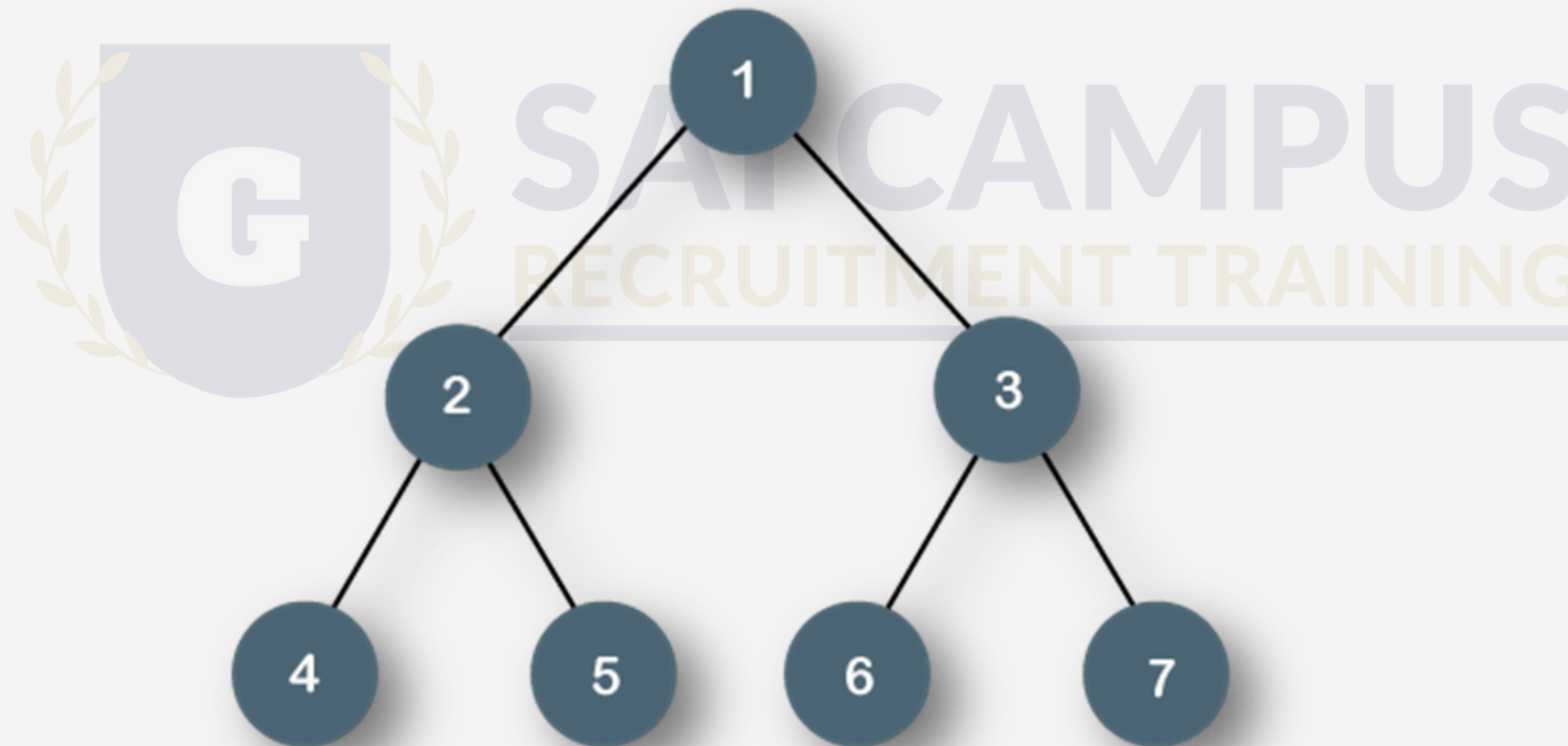
BINARY TREE :

A Tree in which each node can have at most Two child nodes is called a Binary Tree.



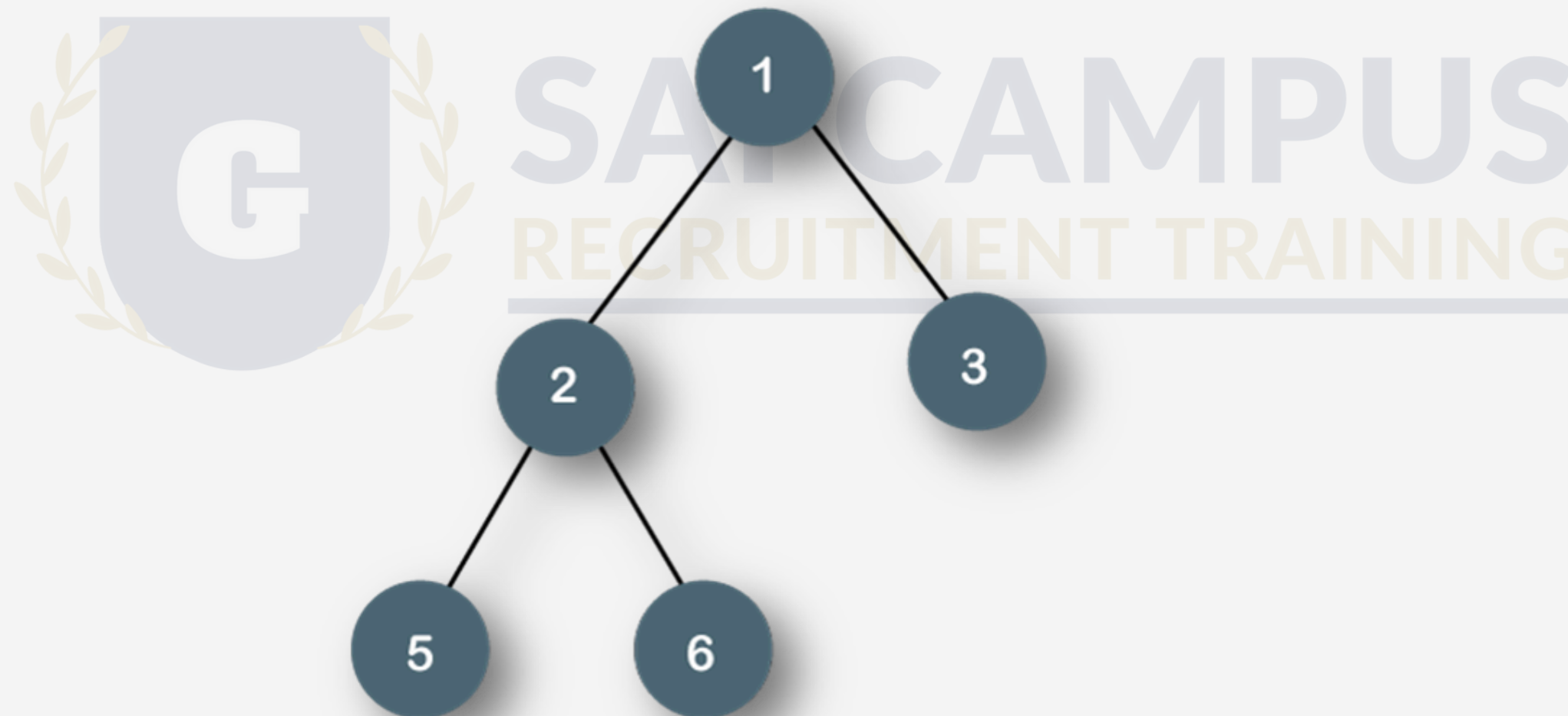
PERFECT BINARY TREE:

A Tree is said to be a Perfect Binary Tree if all its levels are completely fulfilled.



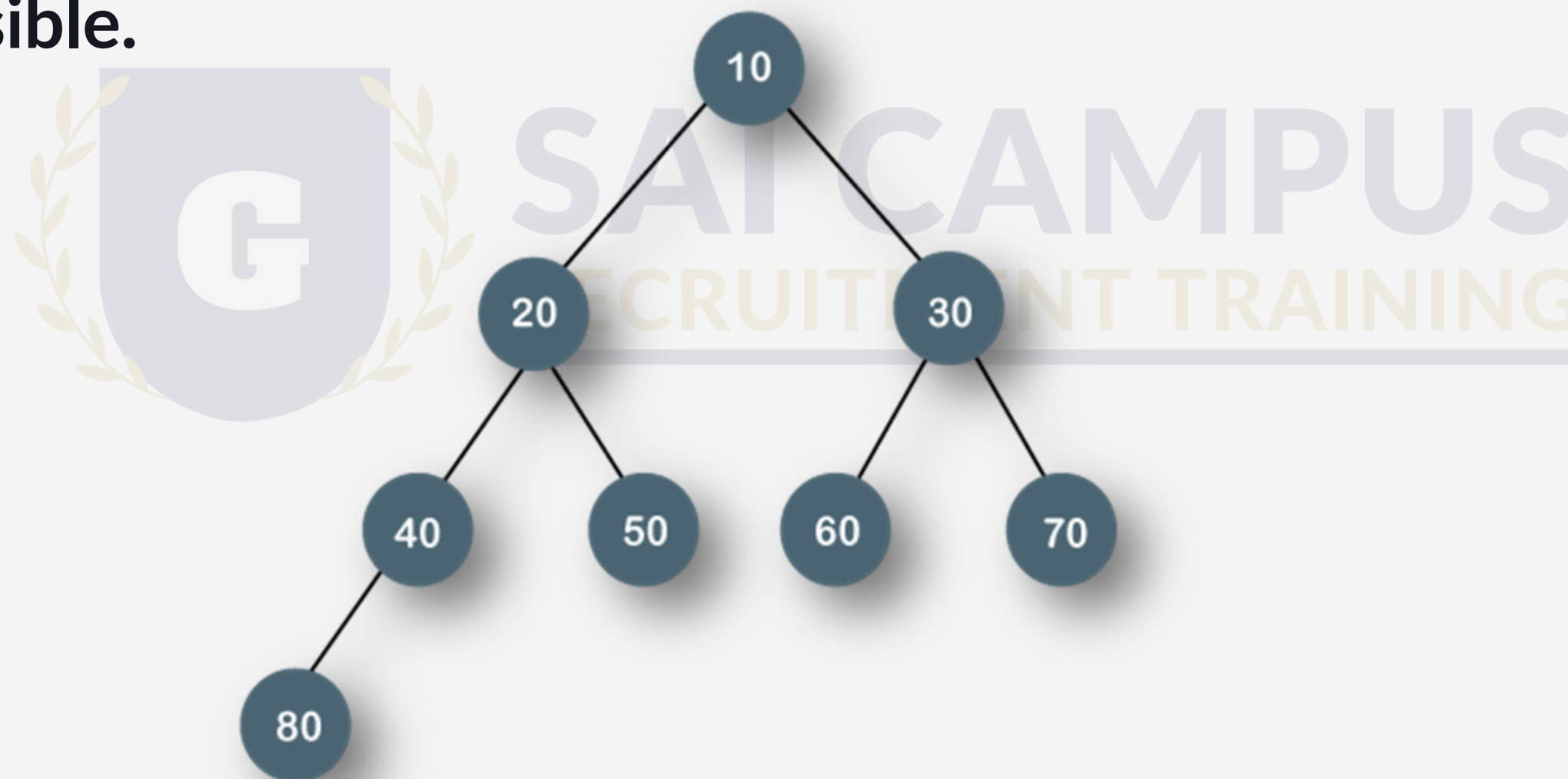
STRICT BINARY TREE:

A Tree is said to be a Strict Binary Tree if all its nodes have either 2 children or does not have children at all.



COMPLETE BINARY TREE:

A Tree is said to be a Complete Binary Tree if all its levels are completely fulfilled or if it is not fulfilled then extra nodes must be as left as possible.



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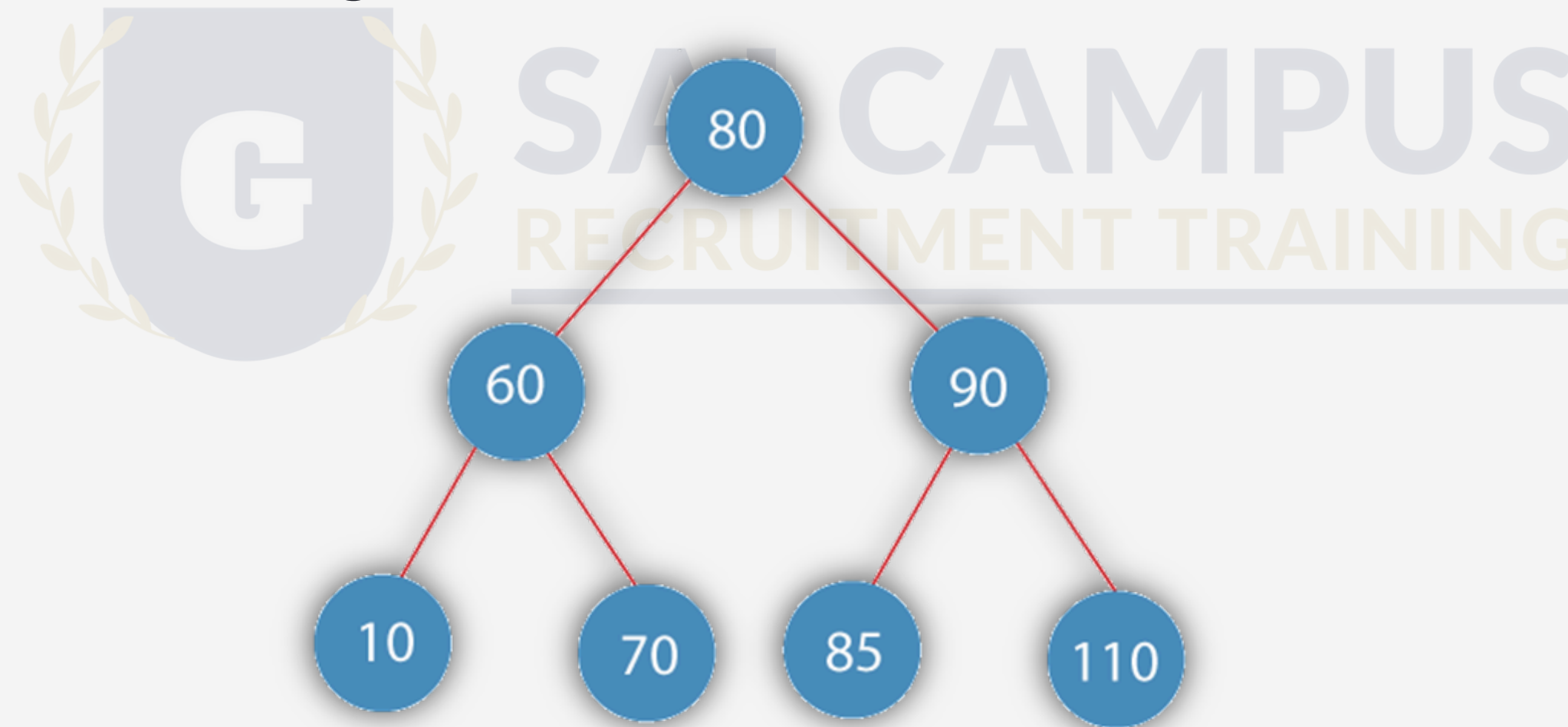
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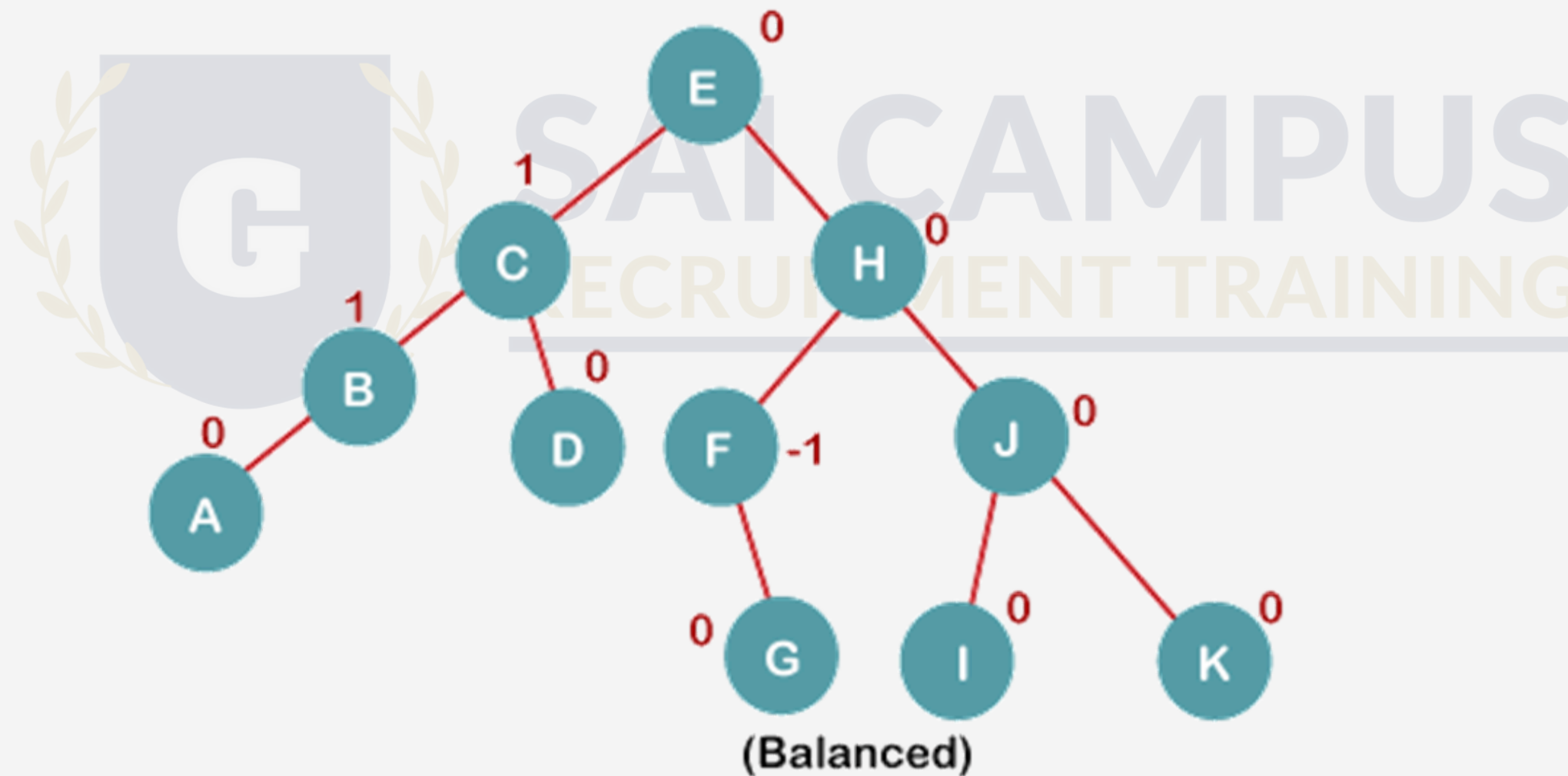
BINARY SEARCH TREE:

A Tree is said to be a Binary Search Tree , if the node with greater value then root goes to its right subtree and if the node with smaller value then root node goes to its left subtree.



AVL TREE:

AVL Tree is a height balancing binary search tree in which the balancing factor of each node is either (0,1,-1).



What is Graph ?

Graph is the collection of vertex and edges. It is denoted by $G(V,E)$.

Example- Google map, Facebook, etc...

TREE	GRAPH
There exists a hierarchical structure, and the top node is called the root node.	The concept of hierarchy leading to a unique root node does not apply here.
Is an acyclic graph.	Cycles can exist.
Must be a connected graph.	Isn't necessarily a connected graph.
Data representation is similar to a tree, with concepts like branches, roots, and leaf nodes.	Data representation is similar to a network
Applications: decision trees, implementing heaps to find max/min numbers, sorting.	Applications: Finding shortest path, navigation, route optimization.



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Sai Campus Recruitment Training is the best institute in Central India for Campus Placements located in Indrapuri, Bhopal. CRT stands for Campus Recruitment Training batch. This is a program where we teach students everything from basics to advance required to get placed in world's top MNC's like Amazon, TCS, Infosys, SAP, Capgemini, Accenture, Cognizant etc. After joining this batch you don't have to worry about anything, We cover every single thing required for your smooth selection. 2000+ students are already placed in their dream MNC's....., YES! you are next!

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ABOUT GAURAV TIWARI SIR:



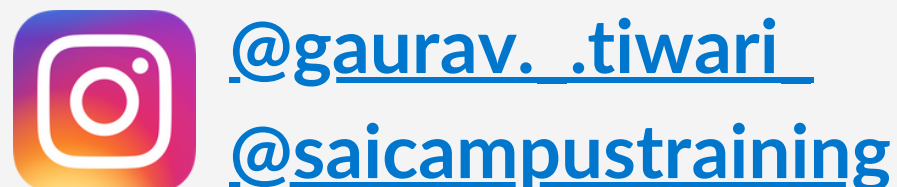
DSA NOTES

Mr. Gaurav Tiwari, a multi-faceted individual with a diverse range of talents and achievements. Mr. Tiwari is widely recognized as a prominent programming trainer, aptitude and logical reasoning instructor, and a proficient communicator. As a famous educator and emerging entrepreneur, he has left an indelible mark in the world of education and business.

Beyond his expertise in training, Mr. Gaurav Tiwari's passion for chess has led him to achieve international recognition as a rated chess player. He has represented himself in numerous national and international chess events, showcasing his strategic acumen and dedication to the game.

As a testament to his innovative abilities, Mr. Tiwari led the team "Ingenious," which emerged victorious in the prestigious Smart India Hackathon 2017, a platform where young minds collaborate to solve real-world challenges through technology.

His outstanding achievements have not gone unnoticed, as he was honored by the Governor of Madhya Pradesh State for securing the highest number of job offers from top multinational companies. This accolade exemplifies his exceptional skills and dedication in the corporate realm.



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Apart from his educational pursuits and entrepreneurial ventures, Mr. Gaurav Tiwari shares his experiences and knowledge with the world as a daily vlogger on his YouTube channel "Gauraverse." Through this platform, he has inspired and motivated thousands of viewers.

In a groundbreaking move, Mr. Tiwari established "Cryptovilla Cafe," India's first cryptocurrency-based cafe, demonstrating his forward-thinking approach to business and technology.

Moreover, his passion for coding and love for teaching have driven him to mentor over 10,000 students, making a lasting impact on the lives and careers of aspiring coders.

Mr. Gaurav Tiwari's remarkable journey and versatile talents make him an extraordinary individual, leaving an indelible mark in the domains of education, entrepreneurship, chess, and technology.



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